ANDREW HAGGSTROM

andrewhaggstrom@gmail.com — (317) 473-3352 — Projects

EDUCATION

Carnegie Mellon, Pittsburgh, Pennsylvania
Master of Science (M.S.) in Robotic Systems Development (MRSD)
Washington University in St. Louis, St. Louis, Missouri
Bachelor's Degree: Biomedical Engineering, Computer Science

WORK EXPERIENCE

Stryker — Mahwah , New Jersey
Business Solutions Engineering Intern — May 2025 – August 2025

- Led daily cross-functional requirements sessions deploying Scrum to translate business needs into technical details.
- Optimized software approval process in Azure DevOps and modeled workflows in Visio to cut down on cycle time and visualize needs to stakeholders.
- Created financial data management pipeline, reducing request time by 20% and increasing request efficiency by 40%.

Medical Artificial Intelligence Lab — St. Louis, Missouri
Undergraduate Researcher — January 2025 – Present

- Examined proteomic Parkinson's data, leveraging machine learning methods to identify predictive protein clusters.
- Applied machine learning libraries including scikit-learn, XGBoost, LightGBM, TensorFlow, and PyTorch.
- Partnered with faculty researchers to analyze biological significance of results and cross-check with prior studies.

MediBeacon — St. Louis , Missouri
Research and Development Intern — May 2024 – August 2024

- Designed molded parts for production using Solidworks and statistical tolerancing methods.
- Automated analysis of safety and verification tests for prototype devices with python and matlab scripts.
- Implemented QSR design controls (820.30) and complied with regulatory standards to attain FDA approval.

Washington University in Saint Louis — St. Louis, Missouri
Problem Solving Team Leader/Academic Mentor — August 2023 – Present

- Directed weekly Biomedical engineering problem-solving sessions focused on CAD skills and material sciences.
- Tutored 20+ undergraduate students in a physics, chemistry, biology, and calculus.

Teaching Assistant for Intro to Machine Learning/Probability and Statistics for Engineers — August 2023 – Present

- Guided students in utilizing machine learning concepts and probability theory through Python coding assignments.

Purdue University Engineering — Indianapolis, Indiana
Biomedical Engineering Intern — June 2023 – August 2023

- Assessed micro-CT, MRI, and NMR scans using MATLAB and deep learning image analysis to monitor bone health.
- Executed ANOVA analysis to support conclusions presented in an abstract for the Orthopedic Research Society.

LEADERSHIP EXPERIENCE

Robotics Club — St. Louis, Missouri
Lead Computer Vision Engineer — August 2025 – Present

- Collaborated with engineers to prototype a wearable assistive device, recognized at the NIH DEBUT Competition.
- Optimized image processing pipelines in Python/OpenCV to improve latency and accuracy of vision-based decisions.
- Integrated cameras with wearable hardware, ensuring communication between vision outputs and haptic feedback.

Engineering Test Kitchen — St. Louis, Missouri
Lead Mechanical Engineer — August 2024 – Present

- Modeled and prototyped 15+ structural components in SolidWorks.
- Directed a 8-person mechanical design team to develop an motorized EV charging system.
- Developed precise CAD models and engineering drawings with GD&T specifications in SolidWorks, leveraging finite element analysis(FEA) to verify structural integrity under load.

Lead Computer Vision Engineer — August 2024 – Present

- Consulted with a local manufacturing company to design an AI/ML solution addressing inventory tracking challenges.
- Built Convolutional neural networks(CNN) in Python to assess pallet images and predict wood piece counts.
- Delivered data-driven recommendations, demonstrating potential for computerized inventory monitoring.

Engineers Without Borders — St. Louis, Missouri
Lead Project Manager — November 2023 – Present

- Spearheaded a collaborative initiative with Missouri River Cities and Towns Initiative, US EPA, and UN Environment Programme to model plastic pollution in the Mississippi River.
- Managed three data analyst teams to model plastic pollution in three neighborhoods in St. Louis.
- Utilized SQL and arcGIS to manipulate collected data and draw meaningful conclusions.

SKILLS

- **Technical:** ROS, Snowflake, Airflow, Spark, SOLIDWORKS, MATLAB(Statistics and Machine Learning Toolbox), Pandas/NumPy, OpenCV, Minitab, GD&T, Deep Learning Specialization Certification from DeepLearning.AI